Portfolio 6: Lists

Student Record Manager

Description

This program allows the user to manage student records using lists.
The user can add student names and grades, view all records, compute the average grade, and identify the highest grade.

Concepts Used

- Lists
- Loops
- Conditionals
- Functions
- Strings
- Numbers

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In [10]:

```
# Portfolio 6: Lists
# Student Record Manager

# Create empty lists
student_names = []
student_grades = []

# Function to display menu
def show_menu():
    print("===== STUDENT RECORDS =====")
    print("1 - Add Student")
    print("2 - View Records")
    print("3 - Compute Average Grade")
    print("4 - Show Highest Grade")
    print("5 - Exit")
```

In [8]:

```
# Create empty lists
student_names = []

math_grades = []
science_grades = []
english_grades = []
```

In [9]:

```
# Function to display menu
def show_menu():
```

```
    print("===== STUDENT RECORDS =====")
    print("1 - Add Student Record")
    print("2 - View Student Records")
    print("3 - Compute Student Average")
    print("4 - Show Highest Average")
    print("5 - Exit")

# Main program loop
while True:

    # Display menu
    show_menu()

    # Ask user choice
    choice = input("Enter your choice: ")

    # Add student record
    if choice == "1":

        name = input("Enter student name: ")

        math_grade = float(input("Enter Mathematics grade: "))
        science_grade = float(input("Enter Science grade: "))
        english_grade = float(input("Enter English grade: "))

        # Save student record
        student_names.append(name)

        math_grades.append(math_grade)
        science_grades.append(science_grade)
        english_grades.append(english_grade)

        print("Student record added successfully!")

    # View all student records
    elif choice == "2":

        if len(student_names) == 0:
            print("No student records found.")

        else:
            print("===== STUDENT RECORDS =====")

            for i in range(len(student_names)):

                print("Student:", student_names[i])

                print("Mathematics:", math_grades[i])
                print("Science:", science_grades[i])
                print("English:", english_grades[i])

    # Compute average grade of one student
    elif choice == "3":

        if len(student_names) == 0:
            print("No student records found.")
```

```
        else:
            search_name = input("Enter student name: ")

            found = False

            for i in range(len(student_names)):

                if student_names[i].lower() == search_name.lower():

                    average = (
                        math_grades[i]
                        + science_grades[i]
                        + english_grades[i]
                    ) / 3

                    print("Student Name:", student_names[i])

                    print("Mathematics:", math_grades[i])
                    print("Science:", science_grades[i])
                    print("English:", english_grades[i])

                    print("Average Grade:", round(average, 2))

                    found = True

            if found == False:
                print("Student not found.")

    # Show highest average
    elif choice == "4":

        if len(student_names) == 0:
            print("No student records found.")

        else:

            highest_average = 0
            top_student = ""

            for i in range(len(student_names)):

                average = (
                    math_grades[i]
                    + science_grades[i]
                    + english_grades[i]
                ) / 3

                if average > highest_average:

                    highest_average = average
                    top_student = student_names[i]

            print("Top Student:", top_student)
            print("Highest Average:", round(highest_average, 2))
```

```python
    # Exit program
    elif choice == "5":

        print("Program ended.")
        break

    # Invalid input
    else:
        print("Invalid choice. Please try again.")
```

```
===== STUDENT RECORDS =====
1 - Add Student Record
2 - View Student Records
3 - Compute Student Average
4 - Show Highest Average
5 - Exit
Student record added successfully!
===== STUDENT RECORDS =====
1 - Add Student Record
2 - View Student Records
3 - Compute Student Average
4 - Show Highest Average
5 - Exit
Student record added successfully!
===== STUDENT RECORDS =====
1 - Add Student Record
2 - View Student Records
3 - Compute Student Average
4 - Show Highest Average
5 - Exit
Student Name: Kian
Mathematics: 90.0
Science: 89.0
English: 95.0
Average Grade: 91.33
===== STUDENT RECORDS =====
1 - Add Student Record
2 - View Student Records
3 - Compute Student Average
4 - Show Highest Average
5 - Exit
Student Name: Theo
Mathematics: 90.0
Science: 95.0
English: 95.0
Average Grade: 93.33
===== STUDENT RECORDS =====
1 - Add Student Record
2 - View Student Records
3 - Compute Student Average
4 - Show Highest Average
5 - Exit
```

```
Top Student: Theo
Highest Average: 93.33
===== STUDENT RECORDS =====
1 - Add Student Record
2 - View Student Records
3 - Compute Student Average
4 - Show Highest Average
5 - Exit
Program ended.
```